

Panel Estimation

EC 421, Set 12

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Prologue

Schedule

Last time

Instrumental variables

Today

Panel estimation

- what panel data buy us,
- fixed effects,
- the difference-in-differences estimator,
- *brief extension*: the synthetic-controls estimator.

Panel estimation

Panel estimation

Introduction

Until now, we mostly used one of two data structures:

Panel estimation

Example

Consider the following panel of six observations:

```
#>      Unit  Year  y_it  x_it
#>   <char> <num> <num> <num>
#> 1:      A     1    10     2
#> 2:      A     2    11     3
#> 3:      A     3    13     4
#> 4:      B     1     7     1
#> 5:      B     2     8     1
#> 6:      B     3     9     2
```

Panel estimation

Example

Consider the following panel of six observations:

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#>   <char> <num> <num> <num>
#> 1:      A     1    10     2
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#> 5:      B     2     8     1
#> 6:      B     3     9     2
```

We observe the same unit repeatedly.

We can compare unit A to itself **over time**, not just A to B (potentially helping with *selection bias*).

Panel estimation

Example

Consider the following panel of six observations:

```
#>      Unit  Year  y_it  x_it
#>   <char> <num> <num> <num>
#> 1:      A     1    10     2
#> 2:      A     2    11     3
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#> 6:      B     3     9     2
```

Similarly, we observe the **period** across **units**.

This feature lets us control for shocks shared across **units** at a given **time**.

Panel estimation

The benefits of panels

Panels help solve (some) econometric problems because they allow us to

Panel estimation

Panels are helpful—not magical

Panel data *can* be helpful for estimating causal effects, but they can still cause problems without careful use.

Panel estimation

The problem with pooled OLS

Suppose you're interested in a simple linear regression model

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}$$

and have **panel data** (with individual i and time t).

Panel estimation

Introducing fixed effects

The solution:

$$y_{it} = \beta_1 x_{it} + \alpha_i + \lambda_t + u_{it}$$

These unit- and period-specific intercepts are called **fixed effects**.

Panel estimation

Fixed effects in action

Q So what are these fixed effects (FE) doing?

Panel estimation

The within transformation

For intuition, focus on unit-level fixed effects.

$$y_{it} = \beta_0 + \beta_1 x_{it} + \alpha_i + u_{it}$$

Panel estimation

Fixed-effects questions

Fixed effects **do not** ask questions like

Why is unit A different from unit B?

In fact, they try to absorb cross-unit differences.

Panel estimation

Example: State minimum wage laws

Suppose we want to estimate the effect of min. wage laws on employment

$$\text{Employment}_{it} = \beta_1(\text{Min. Wage})_{it} + \alpha_i + \lambda_t + u_{it}$$

with a **state** (*i*) by **year** (*t*) panel of employment and min. wage data.

Panel estimation

Example: State minimum wage laws

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$$\text{Employment}_{it} = \beta_1(\text{Min. Wage})_{it} + \alpha_i + \lambda_t + u_{it}$$

with a **state** (*i*) by **year** (*t*) panel of employment and min. wage data.

- α_i absorbs time-invariant, cross-state differences;
- λ_t absorbs national shocks in year *t*.

Q What if some *years* have lower emp. than others (recessions)?

Panel estimation

Example: State minimum wage laws

Suppose we want to estimate the effect of min. wage laws on employment

$$\text{Employment}_{it} = \beta_1(\text{Min. Wage})_{it} + \alpha_i + \lambda_t + u_{it}$$

with a **state** (*i*) by **year** (*t*) panel of employment and min. wage data.

- α_i absorbs time-invariant, cross-state differences;
- λ_t absorbs national shocks in year *t*.

Q So what should we worry about?

Panel estimation

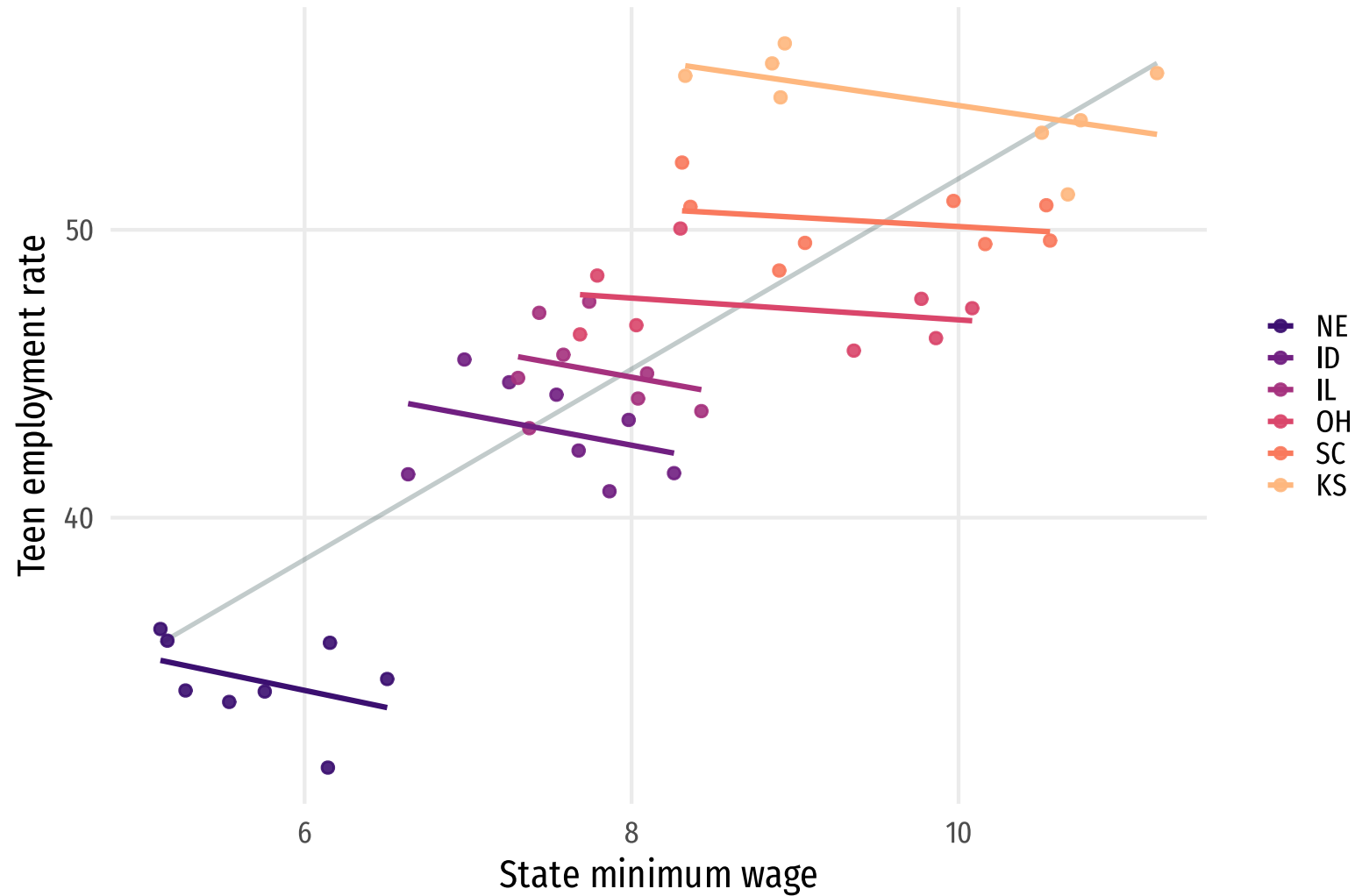
What FE means in that example

In the minimum-wage panel, fixed effects (FEs) net out factors like

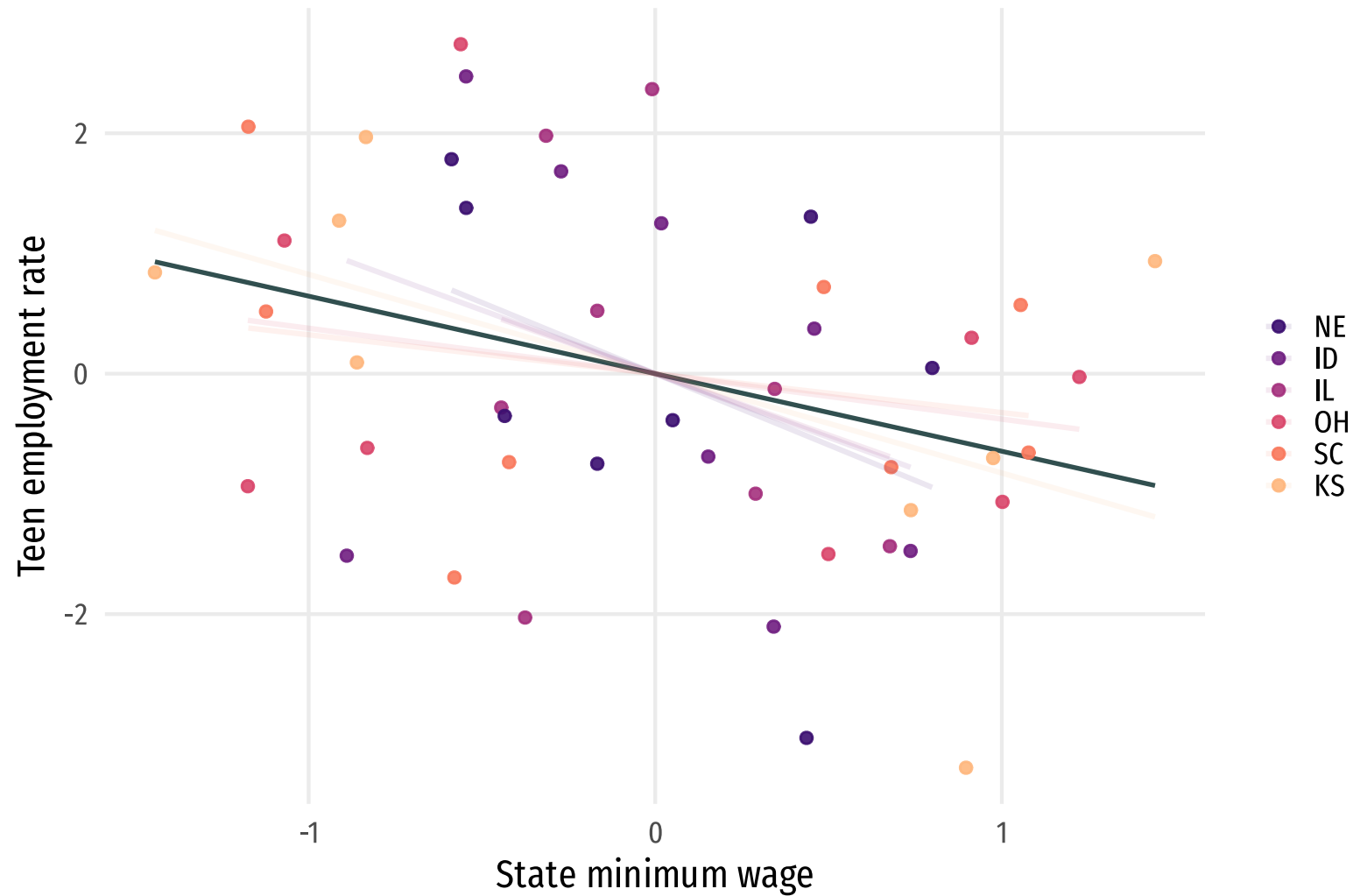
- state geography,
- long-run industry mix,
- baseline labor-market conditions,
- permanent political differences.

Let's see a graphical example.

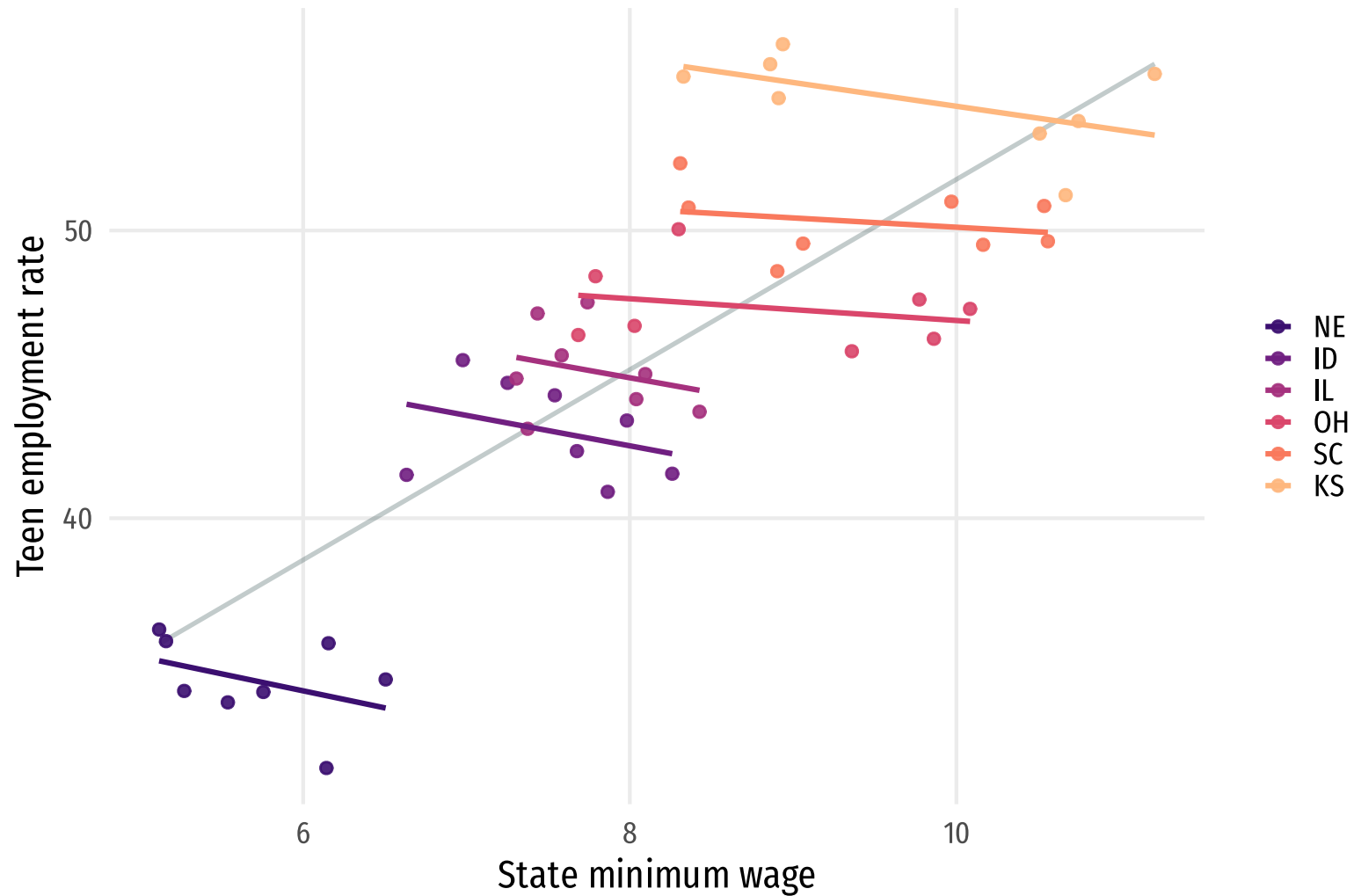
(Simulated data but real FE issues.)



However, the **within-state relationships** tell another story. The FE estimator focuses on this var.—*absorbing* cross-state differences.



Integrating **fixed effects**, the relationship flips.
States that raise their minimum wage see lower teen employment.



This *flipped* relationship is an example of **Simpson's paradox**—when a relationship reverses after controlling for “group”.

Panel estimation

fixest syntax

Now you know the intuition, logic, and mechanics of fixed effects.

You also now understand the `fix` part of the `fixest` package.

To specify FEs: use `lm` syntax along with `|` followed by the FEs.

```
feols(  
  teen_emp ~ min_wage | state + year,  
  cluster = ~ state,  
  data = mw_df  
)
```

Here we have fixed effects for `state` and `year`.

Output

```
#>                                est_ols                est_fe
#> Dependent Var.:                teen_emp            teen_emp
#>
#> Constant                21.3*** (3.00)
#> min_wage                2.93*** (0.345) -1.44*** (0.196)
#> Fixed-Effects:  -----
#> state                    No                    Yes
#> year                     No                    Yes
#> -----
#> S.E.: Clustered          by: state              by: state
#> Observations              320                    320
#> R2                        0.48037                0.96360
```

Output

```
#>                                est_ols                est_fe
#> Dependent Var.:                teen_emp            teen_emp
#>
#> Constant                21.3*** (3.00)
#> min_wage                2.93*** (0.345) -1.44*** (0.196)
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#> R2                       0.48037             0.96360
```

Both $\hat{\beta}_1$ are stat. significant, but they tell **very different stories**.

Output

```
#>                                est_ols                est_fe
#> Dependent Var.:                teen_emp            teen_emp
#>
#> Constant                21.3*** (3.00)
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#> R2                       0.48037                0.96360
```

Q Why is there **no intercept** (β_0) in the FE model?

Output

```
#>                                est_ols                est_fe
#> Dependent Var.:                teen_emp            teen_emp
#>
#> Constant                21.3*** (3.00)
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```

Q What is *within R^2* ? (You may need to scroll down.)

Output

```
#>                                est_ols                est_fe
#> Dependent Var.:                teen_emp            teen_emp
#>
#> Constant                21.3*** (3.00)
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```

The FE model has as high R^2 because fixed effects explain a lot of variation (in essence, means of states and years).

Panel estimation

Summary

Fixed effects help one key problem: **time-invariant omitted variables**.

Difference-in-differences

Core idea

Difference-in-differences¹ (DiD) compares obs. across two dimensions:

1. **Group** comparing

- *treated* individuals (potentially affected by the policy)
- *control* individuals (unaffected by the policy)

2. **Time** comparing

- *before* the policy takes effect (*pre-treatment*)
- *after* the policy takes effect (*post-treatment*)

1. Also called *double-differences* or *diff-in-diff*. Increasingly combined with *two-way fixed effects* (TWFE).

Difference-in-differences

Intuition

DiD functions by taking the difference between two differences:

Ex.

```
#>      Group Period  Mean
#>      <char> <char> <num>
#> 1: Treated    Pre 1.965
#> 2: Control    Pre 0.323
#> 3: Treated    Post 6.172
#> 4: Control    Post 1.036
```

Note You can also write the DiD estimator as two other differences,

$$\begin{aligned}\hat{\tau} &= \Delta \bar{\mathbf{Y}}_{\text{Post}} - \Delta \bar{\mathbf{Y}}_{\text{Pre}} \\ &= [\bar{\mathbf{Y}}(\text{Trt}; \text{Post}) - \bar{\mathbf{Y}}(\text{Ctrl}; \text{Post})] - [\bar{\mathbf{Y}}(\text{Trt}; \text{Pre}) - \bar{\mathbf{Y}}(\text{Ctrl}; \text{Pre})] \\ &= \bar{\mathbf{Y}}(\text{Trt}; \text{Post}) - \bar{\mathbf{Y}}(\text{Ctrl}; \text{Post}) - \bar{\mathbf{Y}}(\text{Trt}; \text{Pre}) + \bar{\mathbf{Y}}(\text{Ctrl}; \text{Pre}) \\ &= [\bar{\mathbf{Y}}(\text{Trt}; \text{Post}) - \bar{\mathbf{Y}}(\text{Trt}; \text{Pre})] - [\bar{\mathbf{Y}}(\text{Ctrl}; \text{Post}) - \bar{\mathbf{Y}}(\text{Ctrl}; \text{Pre})] \\ &= \Delta \bar{\mathbf{Y}}_{\text{Trt}} - \Delta \bar{\mathbf{Y}}_{\text{Ctrl}}\end{aligned}$$

I.e., the change in the treated group (post – pre) relative to the change in the control group.

We're using the control group's **change** as a **counterfactual** for the treated group's **change** had it not received treatment.

Difference-in-differences

Classic example: Card and Krueger (1994)

One famous DiD **application** estimates the **impact of increasing the minimum wage on fast-food employment**.

- *Who:* New Jersey (trt) vs. Pennsylvania (ctrl)
- *When:* before 01 April 1992 (pre) vs. after 01 April 1992 (post)

DiD estimates *(both questions = same answer)*

1. How did NJ-PA empl. diff. change after the policy, relative to before?
2. How did NJ's empl. change after the policy relative to PA's change?

Key assumption:

Difference-in-differences

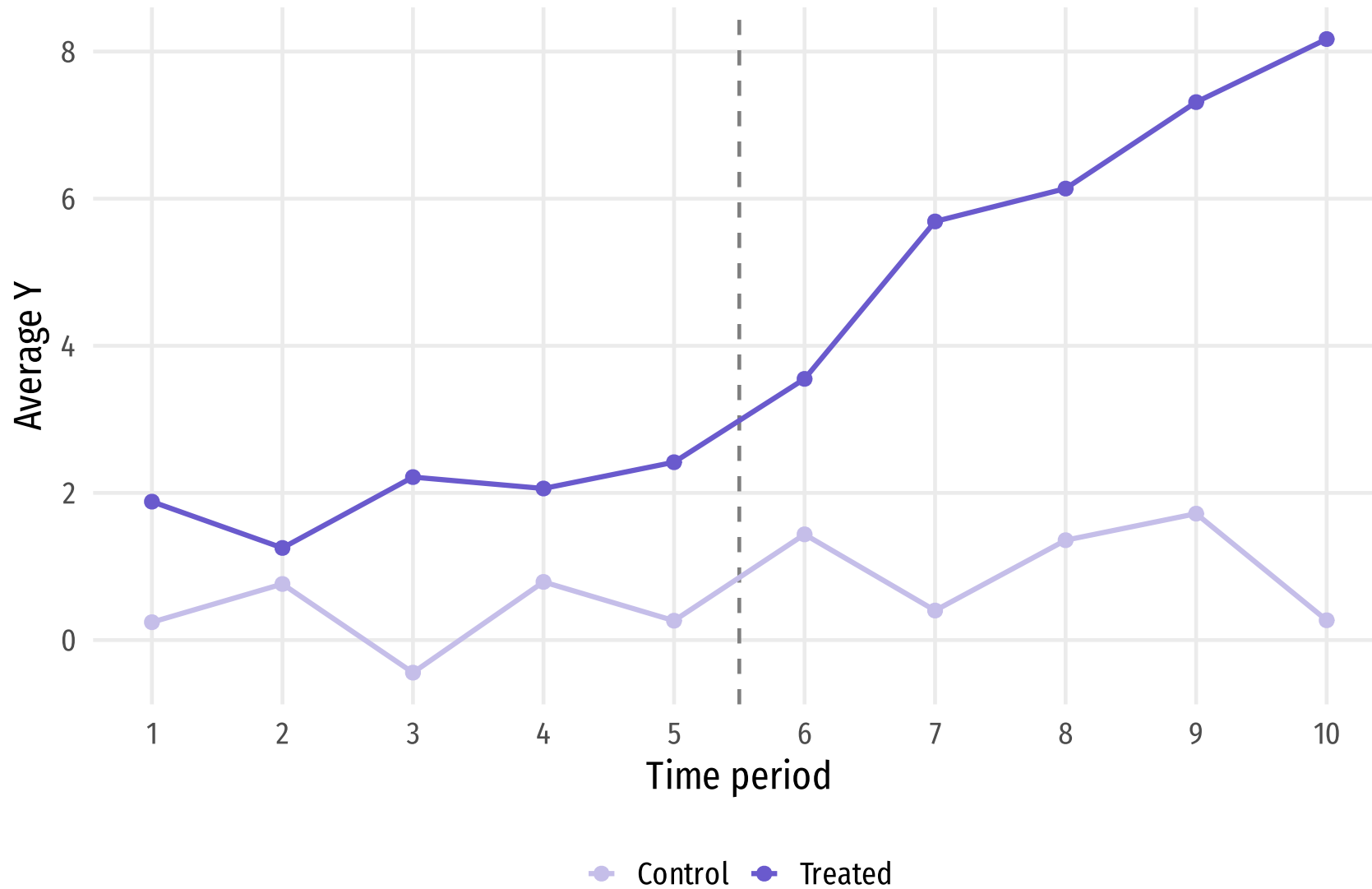
DiD as a regression

With two groups and two periods

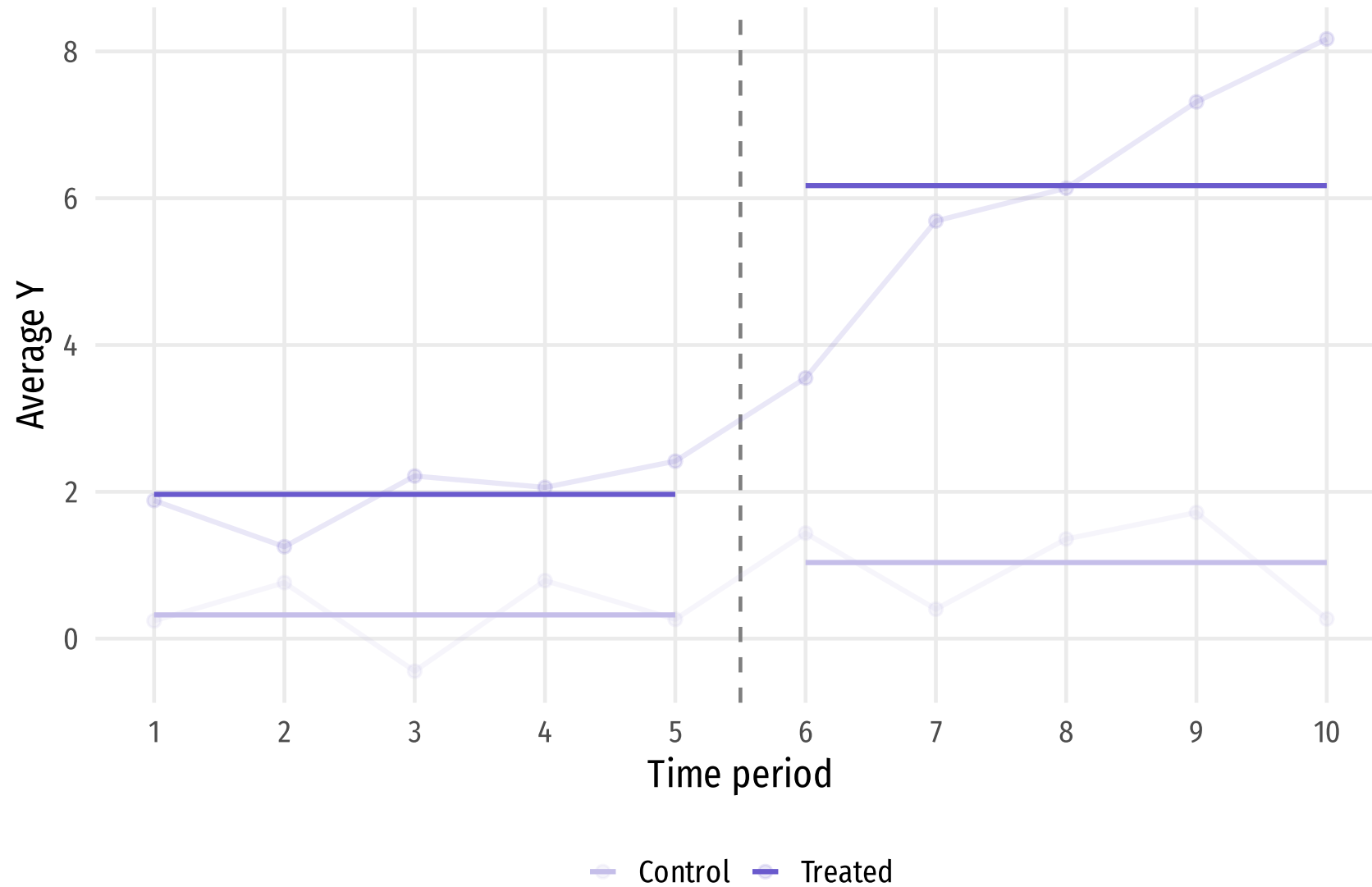
$$y_{it} = \beta_0 + \beta_1 \text{Trt}_i + \beta_2 \text{Post}_t + \delta (\text{Trt}_i \times \text{Post}_t) + u_{it}$$

where Trt_i and Post_t are binary indicators for the treated group and post-treatment period.

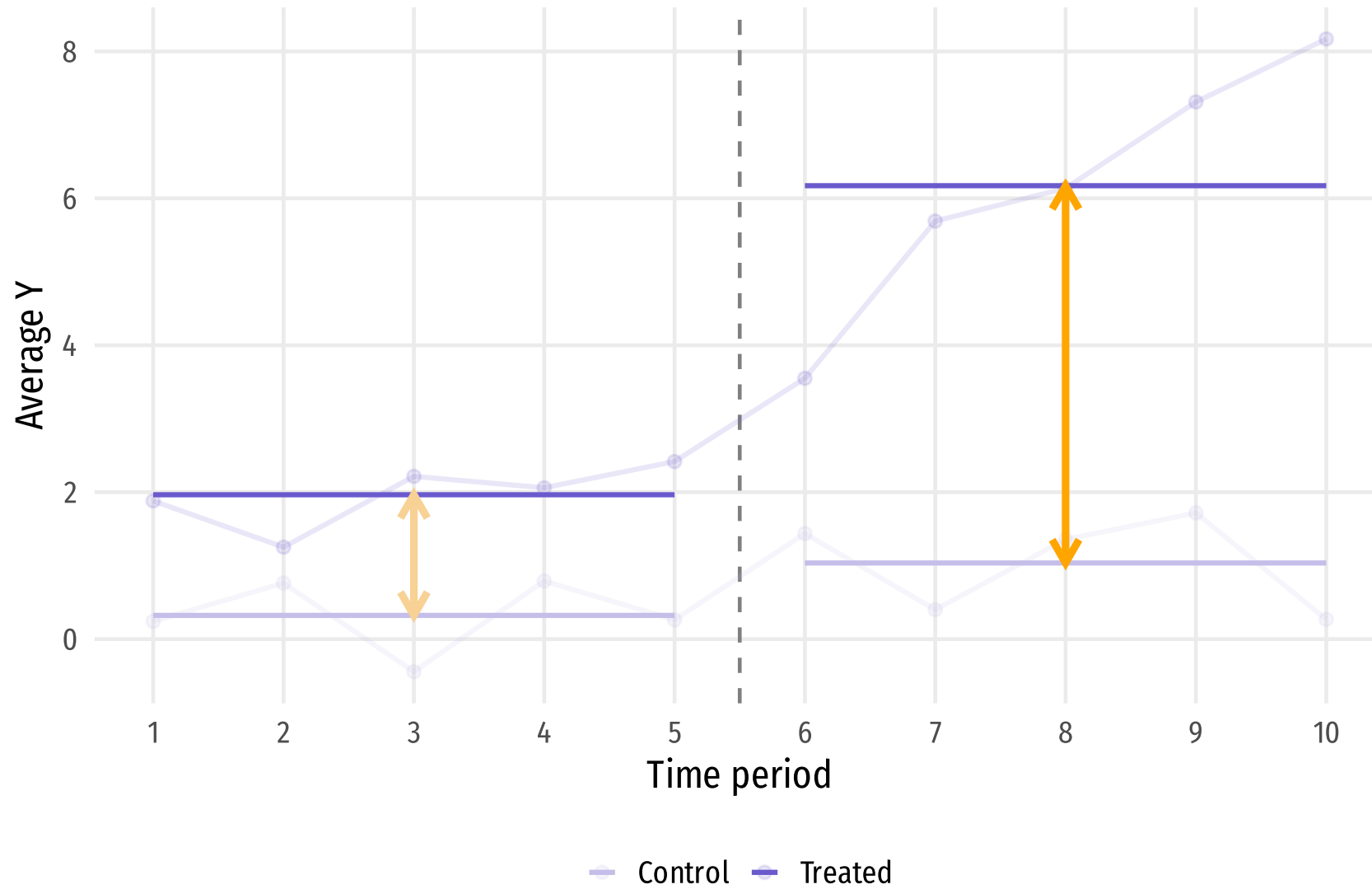
The interaction asks if the post change in the treated group differs from the post change in the control group: the **treatment effect**.



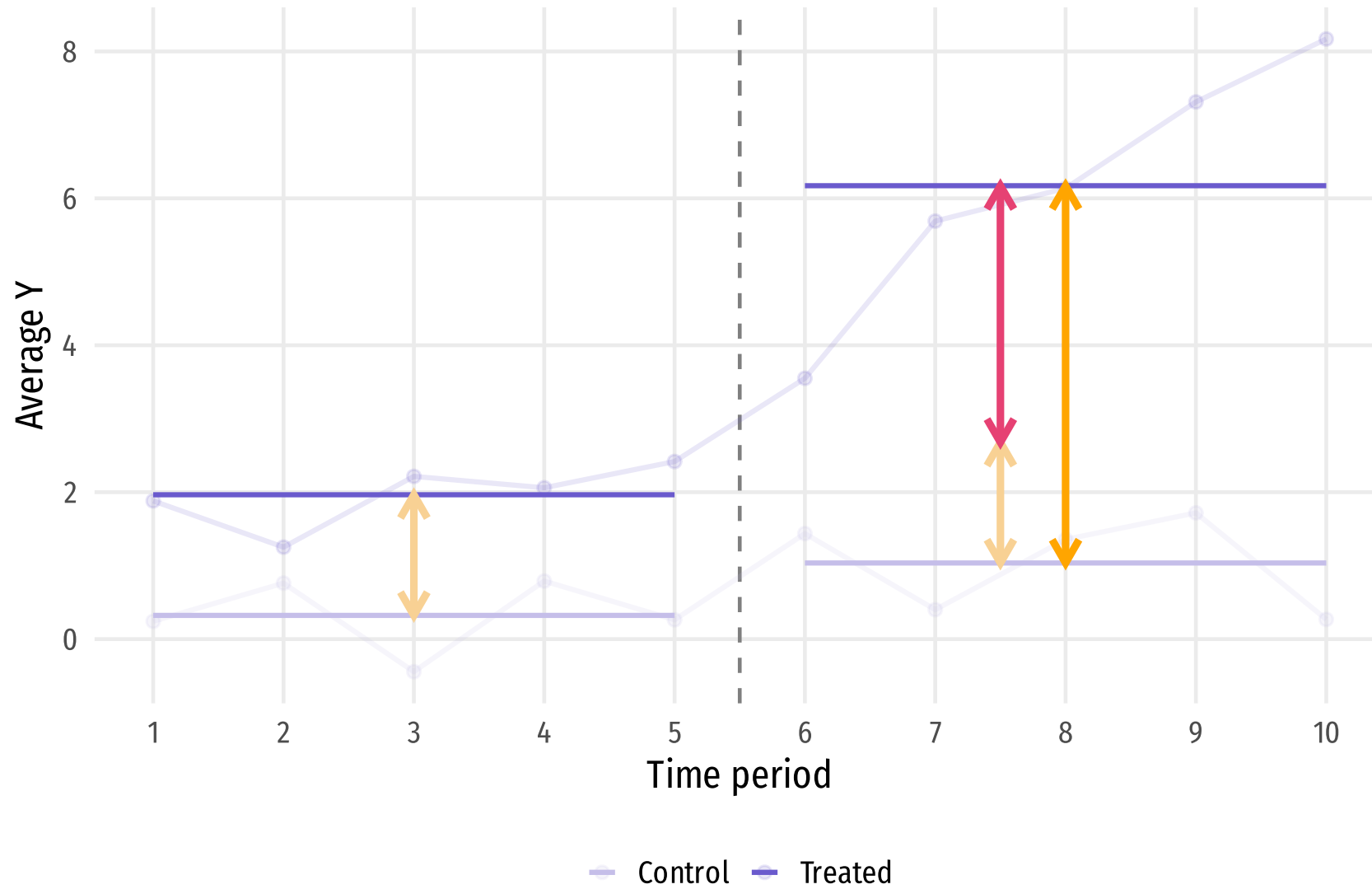
DiD in pictures (*Treatment starts between periods 5 and 6.*)



Here we have **group**-by-**time** means.



The **DiD estimate** compares the **pre-trt** difference to the **post-trt** diff.



The difference in the **pre-trt** and **post-trt** diffs is the **DiD estimate**.

DiD in fixest

```
did_mod =  
  feols(  
    y ~ i(post, treat, ref = 0) | id + period,  
    cluster = 'id',  
    data = base_did  
  )
```

Difference-in-differences

The key DiD assumption

DiD relies upon a critical assumption often called **parallel trends**

In the absence of treatment, treated and control units would have followed the **same average trend**.

Again, we're assuming that the trend of the control units provide a **good counterfactual** for the treated group.

Difference-in-differences

What threatens DiD?

So what violates this *parallel trends* assumption?

Difference-in-differences

DiD's place in the toolbox

DiD is especially useful when

- treatment **timing** is clear,
- **treatment definition** is clear,
- you have good **pre-treatment** data
- **ctrl group** provide a plausible counterfactual for **trt group**,
- the **treatment effect** is tied to a discrete policy or event.

Difference-in-differences

Takeaways

Key concepts from panel estimation, FEs, and DiD:

- **Panel data** let us compare a unit to itself **over time**.
- **Fixed effects** remove **time-invariant omitted variables**.
- $\implies$ **Identifying variation** is **within variation**.
- **DiD** focuses on how trt-ctrl **differences** change over time.

Panels can certainly help, but they still depend on strong assumptions.